

WorldQuant BRAIN Chapter 4 & 5 Master Alpha Guide

Current Score: 7,965 Pts | Goal: 10,000 Pts (GOLD LEVEL ■)

Sovereign Entanglement Asset Systems (SEAS) | Verified: July 26, 2026

1. Chapter 4: Option Alphas & Neutralization Expressions

Alpha Expression	Data Fields Used	Neutralization / Operator	Expected Pass Rate
Option Call-Put IV Skew <code>`group_neutralize(rank(opt_implied_volatility_call - opt_implied_volatility_put), subindustry)`</code>	Call & Put Implied Volatility	Subindustry Neutralization	High Sharpe (> 1.25)
IV vs HV Volatility Premium <code>`group_neutralize(rank(opt_implied_volatility - historical_volatility_20), subindustry)`</code>	Implied Volatility vs Historical Volatility	Subindustry Neutralization	High Fitness (> 1.5)
Densified Bucketed IV Skew <code>`group_neutralize(densify(bucket(rank(opt_implied_volatility_call - opt_implied_volatility_put), range="0.05")), subindustry)`</code>	Call/Put IV Skew + Bucket + Densify	Bucket(), Densify(), Group Neutralization	Submittable Pass

2. Chapter 5: Advanced Multi-Factor & Sentiment Alphas Preview

Alpha Expression	Data Fields Used	Neutralization / Operator	Expected Pass Rate
Analyst Revisions + Volatility <code>`group_neutralize(rank(analyst_target_price_mean / close - 1) * rank(opt_implied_volatility_call), subindustry)`</code>	Analyst Price Target + Option Call IV	Subindustry Neutralization	High Turnover & Sharpe
ESG & Sentiment Trend <code>`group_neutralize(ts_rank(news_sentiment_score, 20), sector)`</code>	News Sentiment + 20-Day TS Rank	Sector Neutralization	Submittable Pass